16.1.2 Introducing GraalVM support for Spring Boot with Spring Native

Spring Native is a new project introduced to support compiling Spring Boot applications with GraalVM. The main goal of Spring Native is to make it possible to compile any Spring application into a native executable using GraalVM without any code changes. To achieve that goal, the project provides an AOT infrastructure (invoked from a dedicated Gradle/Maven plugin) that contributes all the required configurations for GraalVM to AOT-compile Spring classes. The project is one of the latest additions to the Spring portfolio and it's currently in beta. At the time of writing, most Spring libraries are supported, as well as common libraries like Hibernate, Lombok, and gRPC.

For Spring libraries not yet supported, or for your own code, Spring Native offers helpful tools for configuring the GraalVM compiler. For example, if you use reflection or dynamic proxies in your code, GraalVM will require a dedicated configuration to know how to AOT-compile it. Spring Native offers convenient annotations like `@NativeHints` and `@TypedHint` to instruct the GraalVM compiler directly from your Java code, taking advantage of the IDE auto-completion features and type checks.

NOTE Spring Native will come out of the beta phase and become part of the core Spring libraries starting with Spring Framework 6 and Spring Boot 3, which are expected to be released in December 2022.

In this section, we'll explore the features of Spring Native by building Quote Service, a web application that exposes an API to fetch quotes from books.

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